

■ FinTracer AML — Hackathon Winning Playbook

■ The One-Liner (Your Elevator Pitch)

> **FinTracer is a real-time anti-money laundering platform that uses Graph Neural Networks to detect 6 types of fraud patterns in live transaction streams — in under 200ms — and auto-generates FIU-IND compliant reports using Gemini AI.**

Say this in the first 10 seconds. It hits: **real-time**, **GNN**, **6 typologies**, **speed**, **regulatory compliance**, **AI reports**.

■ Presentation Structure (8-10 minutes)

Phase 1: The Problem (1 minute)

> "Indian banks process ■4.5+ crore transactions daily. FIU-IND received 1.7 crore STRs last year. Current rule-based systems catch less than 2% of actual money laundering, while generating 95%+ false positives. Investigators are drowning in noise."

Key stat to drop: "Money laundering costs India an estimated \$150 billion annually — that's 5% of GDP."

The gap: Rule-based systems use static thresholds (e.g., flag anything over ■10 lakh). Criminals just split ■50 lakh into 6 transactions of ■9.5 lakh each. **FinTracer sees the pattern, not just the number.**

Phase 2: Our Innovation — The Architecture (2 minutes)

Show the architecture slide. Walk through **WHY** each layer exists:

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Transaction Stream → Kafka Buffer → Neo4j Graph DB → GNN Inference → Risk Engine → Dashboard

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Gemini AI Reports

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> [!IMPORTANT]

> **Don't just describe the tech stack. Explain WHY you chose each piece.**

| Component | Why This, Not That |

|-----|-----|

| **Neo4j Graph DB** | "Money laundering is fundamentally a **graph problem** — it's about relationships between accounts, not individual transactions. SQL can't do 2-hop neighborhood queries

in 5ms. Neo4j can." |
GNN (not CNN/LSTM)	"Traditional ML treats each transaction independently. Our GIN model sees the **entire subgraph** — it knows that Account A sent to B, B sent to C, and C sent back to A. That circular pattern is invisible to flat models."
Kafka Buffer	"Real banks process 10,000+ TPS. We buffer transactions and batch-flush to Neo4j, handling burst traffic without data loss."
Temperature-Scaled Inference	"Raw neural network outputs are overconfident. We calibrate with temperature=3.0 so our fraud scores are **trustworthy probabilities**, not just classifier outputs."

Phase 3: The 6 Research Innovations (2 minutes)

This is your **differentiator**. Most hackathon teams build one model. You implemented **6 distinct innovations** from recent research papers:

1. Temporal Graph Network (TGN)

> "Transactions have a time dimension. Our TGN doesn't just see **who** transacted with **whom** — it sees **when**. A **50 lakh** transfer at 3 AM to a dormant account scores differently than the same transfer at 2 PM to an active account."

2. Attention-based Temporal Graph Attention (ATGAT)

> "Not all neighbors are equally suspicious. ATGAT assigns **attention weights** — it learns to focus on the most suspicious edges in a subgraph, like an investigator's trained eye."

3. Locally Adaptive Spatial GNN (LAS-GNN)

> "Financial networks aren't uniform. Dense trading hubs look different from sparse rural networks. LAS-GNN adapts its neighborhood aggregation based on **local graph structure**."

4. GARG-AML Structuring Scorer

> "Structuring (smurfing) is the hardest pattern to catch. Criminals split amounts to stay below **10 lakh**. Our dedicated scorer identifies when 5 transactions of **9.5 lakh** are suspiciously similar — even across different time windows."

5. Federated Risk Distillation (Fed-RD)

> "In production, different bank branches have private data. Fed-RD allows branches to **share model improvements**, not raw data — privacy-preserving multi-institutional AML."

6. Explainable AI (XAI) with SHAP/LIME

> "FIU-IND requires investigators to **explain** why a transaction is suspicious. We generate natural language evidence chains and risk factor breakdowns — not just a number."

Phase 4: LIVE DEMO (3 minutes) ■

> [!IMPORTANT]

> **This is where you win or lose.** A working demo beats 100 slides.

Demo Script — Exact Steps:

Step 1: Show the dashboard (15s)

- Open `http://localhost:3000`
- Point to the live metrics, charts, and alert feed
- **"This is our investigator command center. Everything updates in real-time via WebSocket."**

****Step 2: Run the transaction generator (60s)****

- Open a new terminal
- Run: `python transaction\_generator.py --mode manual`
- First, send ****2 legitimate transactions**** (option 1)
- ****Normal salary payments. The model scores these below 0.1 — legitimate.****
- Then send a ****Round-Trip fraud pattern**** (option 5)
- ****Now watch. I'm sending a round-trip — A sends to B, B to C, C back to A. Same amount, 30 minutes apart.****
- ****Switch to dashboard**** — show the toast notification appearing in real-time
- ****Detected in under 200ms. Risk score 0.82. The GNN saw what a rule-based system never would.****

****Step 3: Show FlowScope graph (30s)****

- Click the detected alert → FlowScope
- ****Here's the actual fund flow network. Red nodes = high risk. The animated dots show money movement direction. You can see the circular pattern — that's the round-trip.****
- Click a node → show account details
- Click an edge → show transaction details

****Step 4: Show the evidence chain (20s)****

- Go to Alerts tab → expand the alert
- ****Every detection comes with a full evidence chain. Step-by-step, with timestamps, amounts, and per-edge suspicion scores. This is what an investigator needs.****

****Step 5: Generate an STR report (30s)****

- Go to Reports tab → click Generate
- ****One click generates an FIU-IND compliant Suspicious Transaction Report. Gemini AI writes the narrative. An investigator would spend 2 hours on this. We do it in 3 seconds.****

****Step 6: Show Risk Breakdown (15s)****

- Expand any alert → Risk Breakdown tab
- ****Full transparency. Base GNN score, amount factor, velocity factor, night factor, account age factor. The investigator can see exactly WHY the model flagged this.****

Phase 5: Production Readiness & Impact (1 minute)

> "This isn't a prototype. We've architected for production:"

- ****Kafka with replication factor 3**** — zero transaction loss
- ****Neo4j Causal Clustering**** — concurrent reads for parallel detection engines
- ****Continual learning**** — the model retrains nightly with 70% new + 30% historical data
- ****Mutual TLS**** for cross-branch federated learning
- ****Full audit trail**** — every SHAP output is stored immutably in PostgreSQL

****Impact numbers to quote:****

- ****Current systems: 95% false positive rate. FinTracer: targeting <15%****
- ****Investigation time per alert: 2 hours → 15 minutes****
- ****Time to detection: batch processing (hours) → real-time (200ms)****

Phase 6: Closing (30 seconds)

> "Money laundering funds terrorism, drug trafficking, and human trafficking. Every hour of detection delay means money moves further from reach. FinTracer doesn't just detect faster — it detects **smarter**, using the same graph structures that criminals use to hide. We don't fight patterns with rules. We fight patterns with patterns."

■ Judge Q&A; — Anticipated Questions & Winning Answers

Q: "How is this different from existing AML tools like NICE Actimize or SAS?"

> "Those are rule-based with ML bolted on. They flag single transactions against static thresholds. We're **graph-native** — our fundamental unit isn't a transaction, it's a **subgraph**. When we score a transaction, we look at its entire 2-hop neighborhood within a 72-hour window. That's why we catch round-trips, fan-outs, and structuring that rule-based systems miss entirely."

Q: "What's your false positive rate?"

> "On our synthetic dataset with balanced typologies, our GIN model achieves an F1-score across all 7 classes with temperature-calibrated confidence. In practice, the risk engine applies contextual multipliers — amount, velocity, time-of-day, account age — that further reduce false positives. Our target for production is <15% FPR, compared to industry standard of 95%+."

Q: "Can this handle real bank transaction volume?"

> "Yes. The Kafka buffer handles burst traffic by batching transactions before Neo4j write. The FastAPI inference service runs at ~200ms per subgraph on CPU. With GPU (NVIDIA Triton), that drops to <50ms. We've designed for horizontal scaling with Kubernetes StatefulSets and Neo4j read replicas."

Q: "How do you handle privacy / data protection?"

> "Three layers. First, the preprocessing layer strips PII before sending data to the ML engine. Second, Fed-RD enables cross-branch model updates via gradient sharing, never raw data. Third, all data at rest is encrypted, and all microservice communication uses TLS 1.3."

Q: "What if criminals adapt to your patterns?"

> "That's exactly why we built continual learning. The GARG-AML replay buffer stores 30% historical patterns and retrains nightly. As new fraud patterns emerge, the model evolves. The TGN architecture specifically encodes temporal dynamics — it learns that patterns shift over time."

Q: "Why Neo4j instead of a regular database?"

> "Show them the Cypher query. A 2-hop neighborhood query with time filtering across millions of edges takes ~5ms in Neo4j. The equivalent SQL query with self-joins would take 10-100x longer and be unmaintainable. Money laundering IS a graph problem — we use a graph database."

Q: "Is the Gemini report generation reliable?"

> "We use Gemini as an accelerator, not a dependency. Every generated narrative must cite specific transaction IDs, amounts, and timestamps from the actual evidence chain. If Gemini is unavailable, the system falls back to rule-based narratives. The facts always come from our detection engine, never hallucinated."

■ Power Moves to Stand Out

1. Live Terminal Split-Screen

Run the transaction generator in one terminal, have the dashboard open in the browser. Judges see transactions flow in real-time. **Nothing beats a live demo.**

2. The "Smurfing" Moment

> Send structuring transactions: **"I'm sending 5 payments of ₹9.5 lakh each. Every single one is below the ₹10 lakh reporting threshold. A rule-based system would mark all 5 as legitimate. Watch what FinTracer does..."**

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> Dashboard shows: **Structuring detected, Risk: HIGH**

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> **"It saw the PATTERN. Not the number."**

This is your mic-drop moment.

3. Show the Code, Not Just the Demo

If judges ask about technical depth, pull up:

- ``inference_service.py`` line 180-228: The GIN model architecture
- ``inference_service.py`` line 412-528: The risk engine with 5 contextual factors
- ``Neo4jRepository.java`` line 452-561: The 2-hop Cypher neighborhood query
- The Kafka buffering in ``TransactionService.java``: batch flush + re-queue on failure

4. Use Indian Context

- Reference **FIU-IND** (not FinCEN)
- Use **PMLA 2002** (not BSA)
- Currency in **₹ INR** (not USD)
- Payment methods: **NEFT, RTGS, IMPS, UPI** (not SWIFT, ACH)
- This shows you understand the regulatory landscape, not just the tech

■ ■ Pre-Demo Checklist

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- Neo4j Desktop running (bolt://localhost:7687)
- Docker Compose up (Kafka + Zookeeper)
- Spring Boot backend running (:8080)
- Python inference service running (:8000)
- Frontend dev server running (:3000)
- Transaction generator ready (manual mode)
- Browser on dashboard page, zoomed to 90% for better demo visibility
- Terminal positioned next to browser for split-screen effect

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Startup Order:

1. ``docker-compose up -d`` (Kafka)
2. Start Neo4j Desktop
3. ``cd backend/backend && ./mvnw spring-boot:run``
4. ``cd inference && python inference_service.py``
5. ``cd frontend && npm run dev``
6. Open ``http://localhost:3000``

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7. `cd transaction && python transaction_generator.py --mode manual`
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■ If Time is Short (5-minute version)

1. **Problem** (30s): "■150B laundered annually. Rule-based systems fail."
2. **Solution** (30s): "Graph Neural Networks on real-time transaction streams."
3. **LIVE DEMO** (3 min): Send legit → send fraud → show detection → show graph → show report
4. **Close** (1 min): "6 research innovations, production-ready architecture, FIU-IND compliant."

> [!TIP]

> **The demo IS your presentation.** Minimize slides. Maximize the live system. Judges remember what they SAW, not what they heard.